No.	Status	Date
LNDE-RPT-01-037-DS-4-RA-104	Issued	2001-03-02

UT Thickness Data Sheet - 4-RA-104

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 4-RA-104, located within the ATU-100. The scope of work involves the assessment of the rich side of the L/R exchanger to determine current wall thickness and evaluate asset integrity.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Basis

The inspection was performed under the requirements of QSCP-NDT-001 Rev 0, the site NDE procedure for ultrasonic thickness measurement within the ATU-100. All measurements were conducted in accordance with the standard ASTM E797.

The data presented herein supports the findings detailed in LNDE-RPT-01-037, which covers the 5-year Class 1 survey for sour gas, regenerator overhead, and rich amine piping.

Component Specifications

Line 4-RA-104 is constructed from A106 Gr.B with a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design limits for this line are 862 kPa(g) and 127 °C.

The minimum allowable thickness (t-min) for this component is 3.81 mm. Any reading below this threshold is subject to an NCR.

Integrity Assessment

Corrosion rates are calculated based on the larger of the long-term rate since the first reading or the short-term rate over the latest pair of readings at least 3 years apart. Remaining life is determined by the formula (t - t-min) / rate.

The next thickness reading interval is set at no more than half the remaining life, constrained by the Class 1 maximum of 5 years for thickness and external visual inspections.

Thickness readings, measured 2001-03-02

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	1996-03-08	7.12	6.82	0.060	0.060	>50	2006-03-02	
6-SG-101-02	7.11	3.56	1996-03-08	7.05	6.71	0.068	0.068	46.2	2006-03-02	
6-SG-101-03	7.11	3.56	1996-03-08	7.08	6.74	0.068	0.068	46.6	2006-03-02	
6-SG-101-04	7.11	3.56	1996-03-08	7.09	6.73	0.072	0.072	43.9	2006-03-02	
6-SG-101-05	7.11	3.56	1996-03-08	7.14	6.68	0.092	0.092	33.8	2006-03-02	
6-SG-101-06	7.11	3.56	1996-03-08	7.07	6.87	0.040	0.040	>50	2006-03-02	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	1996-03-08	7.14	6.72	0.084	0.084	37.5	2006-03-02	
6-SG-102-02	7.11	3.56	1996-03-08	7.11	6.72	0.078	0.078	40.4	2006-03-02	
6-SG-102-03	7.11	3.56	1996-03-08	7.04	6.69	0.070	0.070	44.6	2006-03-02	
6-SG-102-04	7.11	3.56	1996-03-08	6.99	6.64	0.070	0.070	43.8	2006-03-02	
6-SG-102-05	7.11	3.56	1996-03-08	7.13	6.93	0.040	0.040	>50	2006-03-02	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	1996-03-08	8.52	8.39	0.026	0.026	>50	2011-03-02	
4-RA-101-02	8.56	3.81	1996-03-08	8.54	8.45	0.018	0.018	>50	2011-03-02	
4-RA-101-03	8.56	3.81	1996-03-08	8.61	8.42	0.038	0.038	>50	2011-03-02	
4-RA-101-04	8.56	3.81	1996-03-08	8.63	8.37	0.052	0.052	>50	2011-03-02	
4-RA-101-05	8.56	3.81	1996-03-08	8.60	8.42	0.036	0.036	>50	2011-03-02	
4-RA-101-07	8.56	3.81	1996-03-08	8.60	8.51	0.018	0.018	>50	2011-03-02	
4-RA-101-08	8.56	3.81	1996-03-08	8.68	8.30	0.076	0.076	>50	2011-03-02	
4-RA-101-09	8.56	3.81	1996-03-08	8.57	8.43	0.028	0.028	>50	2011-03-02	
4-RA-101-10	8.56	3.81	1996-03-08	8.67	8.48	0.038	0.038	>50	2011-03-02	
4-RA-101-11	8.56	3.81	1996-03-08	8.68	8.39	0.058	0.058	>50	2011-03-02	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	1996-03-08	8.54	8.48	0.012	0.012	>50	2011-03-02	
4-RA-102-02	8.56	3.81	1996-03-08	8.57	8.50	0.014	0.014	>50	2011-03-02	
4-RA-102-03	8.56	3.81	1996-03-08	8.56	8.37	0.038	0.038	>50	2011-03-02	
4-RA-102-04	8.56	3.81	1996-03-08	8.54	8.35	0.038	0.038	>50	2011-03-02	
4-RA-102-05	8.56	3.81	1996-03-08	8.55	8.36	0.038	0.038	>50	2011-03-02	
4-RA-102-06	8.56	3.81	1996-03-08	8.51	8.38	0.026	0.026	>50	2011-03-02	
4-RA-102-07	8.56	3.81	1996-03-08	8.58	8.49	0.018	0.018	>50	2011-03-02	
4-RA-102-08	8.56	3.81	1996-03-08	8.49	8.36	0.026	0.026	>50	2011-03-02	
4-RA-102-09	8.56	3.81	1996-03-08	8.55	8.35	0.040	0.040	>50	2011-03-02	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	1996-03-08	11.00	10.82	0.036	0.036	>50	2011-03-02	
6-RA-103-02	10.97	4.57	1996-03-08	11.09	10.69	0.080	0.080	>50	2011-03-02	
6-RA-103-03	10.97	4.57	1996-03-08	10.99	10.84	0.030	0.030	>50	2011-03-02	
6-RA-103-04	10.97	4.57	1996-03-08	10.98	10.59	0.078	0.078	>50	2011-03-02	
6-RA-103-05	10.97	4.57	1996-03-08	10.95	10.61	0.068	0.068	>50	2011-03-02	
6-RA-103-06	10.97	4.57	1996-03-08	11.01	10.86	0.030	0.030	>50	2011-03-02	
6-RA-103-07	10.97	4.57	1996-03-08	10.99	10.79	0.040	0.040	>50	2011-03-02	
6-RA-103-08	10.97	4.57	1996-03-08	10.97	10.83	0.028	0.028	>50	2011-03-02	
6-RA-103-09	10.97	4.57	1996-03-08	11.01	10.85	0.032	0.032	>50	2011-03-02	
6-RA-103-10	10.97	4.57	1996-03-08	11.02	10.78	0.048	0.048	>50	2011-03-02	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	1996-03-08	7.59	7.26	0.066	0.066	>50	2006-03-02	
3-RV-101-02	7.62	3.56	1996-03-08	7.57	7.31	0.052	0.052	>50	2006-03-02	
3-RV-101-03	7.62	3.56	1996-03-08	7.52	7.15	0.074	0.074	48.3	2006-03-02	
3-RV-101-04	7.62	3.56	1996-03-08	7.59	7.18	0.082	0.082	44.0	2006-03-02	
3-RV-101-05	7.62	3.56	1996-03-08	7.67	7.25	0.084	0.084	43.8	2006-03-02	
3-RV-101-06	7.62	3.56	1996-03-08	7.59	7.20	0.078	0.078	46.5	2006-03-02	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	1996-03-08	5.53	5.29	0.048	0.048	>50	2006-03-02	
2-RV-102-02	5.54	2.79	1996-03-08	5.59	5.12	0.094	0.094	24.7	2006-03-02	
2-RV-102-03	5.54	2.79	1996-03-08	5.48	5.24	0.048	0.048	>50	2006-03-02	
2-RV-102-04	5.54	2.79	1996-03-08	5.58	5.33	0.050	0.050	>50	2006-03-02	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	1996-03-08	8.51	8.40	0.022	0.022	>50	2011-03-02	
4-RA-104-02	8.56	3.81	1996-03-08	8.55	8.41	0.028	0.028	>50	2011-03-02	
4-RA-104-03	8.56	3.81	1996-03-08	8.54	8.46	0.016	0.016	>50	2011-03-02	
4-RA-104-04	8.56	3.81	1996-03-08	8.59	8.35	0.048	0.048	>50	2011-03-02	
4-RA-104-05	8.56	3.81	1996-03-08	8.60	8.29	0.062	0.062	>50	2011-03-02	
4-RA-104-06	8.56	3.81	1996-03-08	8.59	8.39	0.040	0.040	>50	2011-03-02	
4-RA-104-07	8.56	3.81	1996-03-08	8.59	8.29	0.060	0.060	>50	2011-03-02	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	2001-03-02	_____